

Enhancing Infrared-Laser Dissociation of Molecules with the Electromagnetic Vacuum

Johan F. Triana^{*}

Departamento de Física, Universidad Católica del Norte, Avenida Angamos 0610, Antofagasta, Chile

Felipe Herrera[†]

Departamento de Física, Universidad de Santiago de Chile, Av. Victor Jara 3493, Santiago, Chile and Millennium Institute for Research in Optics, Concepción, Chile



(Received 24 November 2025; accepted 15 May 2026; published 7 July 2026)

Controlling bond breaking is a long-standing goal in molecular physics. Infrared nanocavities are currently being developed for reaching exotic coupling regimes of cavity QED with a few molecules, but it is not well understood how chemical reactions would proceed in such systems. We study infrared laser photodissociation of a single molecule that strongly interacts with a resonant infrared vacuum, subject to a strong laser field that either resonantly drives the molecule or injects photons into the cavity. We show that the intensities required for photodissociation are significantly lower inside the cavity than in free space. By directly injecting photons into the cavity, the molecule dissociates with 2 orders of magnitude less laser energy than by directly driving the vibrational mode. This photodissociation enhancement is a purely quantum mechanical effect that cannot be captured semiclassically. Our Letter provides fundamental mechanistic understanding of chemical dynamics that can be used for designing new types of nanophotonics experiments that probe single-molecule chemistry.

DOI: [10.1103/s2zc-6lxs](https://doi.org/10.1103/s2zc-6lxs)

Controlling chemical reactions using electromagnetic fields is a long-standing goal in chemistry and physics [1]. Infrared cw lasers can be used to resonantly drive the fundamental vibration frequency to excite the lowest vibrational states of a molecule, initiating chemical processes that involve bond breaking [2–4]. However, infrared photodissociation is challenging to control due to the anharmonic nature of vibrational energy levels, which tend to become far detuned from the laser frequency for highly excited levels. Overcoming this anharmonic blockade effect relies on multiphoton vibrational excitation using strong laser fields [5–7], negatively chirped pulses [8–11], and optimized pulse shaping [12–14]. In polyatomic molecules, the spectral anharmonicity can be compensated with a higher density of states at intermediate regions of the vibrational potential, which results from the interaction with other vibrational modes of the molecule [15–17]. The higher density of states produces a vibrational quasicontinuum that facilitates ladder climbing to higher vibrational levels, reducing the laser intensity needed to break a molecular bond [17].

It is unknown how these laser chemistry concepts generalize when molecules are inside infrared cavities in the strong coupling regime of cavity QED. Ladder climbing processes should then occur through hybrid light-matter states, known as vibrational polaritons [18–21], which have

been experimentally observed with molecular ensembles in Fabry-Perot cavities [22–24] and nanophotonic devices [25–29]. Vibrational polaritons have been associated with observed modifications of chemical reactivity in infrared cavities [30–32] and plasmonic resonators [33]. Recently, single and few-molecule infrared nanocavities have emerged as promising light-matter interaction platforms with ultrasmall cavity mode volumes that enable strong [34] and ultrastrong [35,36] coupling. Modifications of the photodissociation process in nanocavity systems can be expected based on spectral modification of radiative heat transfer rates [33,37], and fundamental changes to vibrational ladder climbing due to cavity vacuum effects are expected for single molecules in ultrastrong coupling [38], but the combined influence of strong laser excitation with strong vacuum coupling effects has not been explored.

In this Letter, we study the influence of the infrared nanocavity vacuum on the vibrational ladder-climbing process of an individual nonpolar polyatomic molecule in strong coupling. Using carbon disulfide (CS₂) as an example [39], we show that vacuum-assisted infrared photodissociation leads to higher dissociation yields with respect to free space at equal driving energy. We demonstrate that this enhancement of photodissociation probability is a purely quantum mechanical effect by comparing with semiclassical mean-field models. We also show that injecting photons directly into the nanocavity field is the most energetically efficient mechanism for breaking

^{*}Contact author: johan.triana@ucn.cl

[†]Contact author: felipe.herrera.u@usach.cl

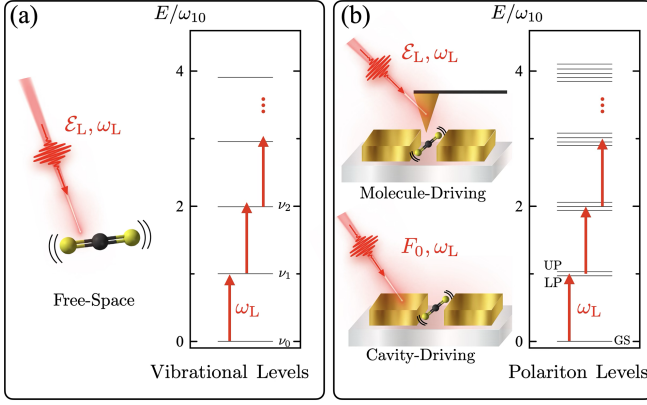


FIG. 1. Driven molecule-cavity coupled system. (a) CS_2 molecule in free space driven by an infrared laser. Multi-photon transitions between vibrational levels lead to dissociation. (b) CS_2 in a strongly confined infrared electromagnetic vacuum generated by metallic nanostructures. Multiphoton infrared absorption occurs between polariton levels. In the molecule-driving case, the laser drives a nanotip to excite the molecule, and in the cavity-driving scenario, the cw laser injects photons directly on the near field of the coupled system.

molecular bonds with infrared laser fields resonant with the fundamental vibrational frequency.

In free space, the simplest type of photodissociation is done with a continuous wave (cw) monochromatic laser that is resonant with the fundamental vibrational frequency [as Fig. 1(a) illustrates] [40,41]. Since molecular vibrations are anharmonic, transitions between high-energy levels are suppressed, which is usually compensated by increasing the laser intensity or via frequency chirping. Modern implementations of infrared photodissociation use a combination of strategies to optimize the dissociation yields [10–13,37].

In a cavity, the laser drives transitions between polariton eigenstates in two ways: through the electric dipole of the molecule ($\hat{d}$) or the quadrature field of the cavity ($\hat{x}$). The dynamics of the ladder climbing process that results in bond-breaking occurs along the high density of polariton states that form a quasicontinuum, similar to the vibrational quasicontinuum established in polyatomic molecules [15–17]. This high density of states enhances the ladder-climbing mechanism relative to free space, increasing dissociation yields. Photodissociation enhancement occurs because more states are accessible in comparison with the bare vibrational spectrum, enabling matter-matter, light-light, and light-matter transitions at infrared frequencies [21,42]. These processes extend established laser-assisted chemistry concepts to cavity-QED conditions.

We describe the C-S stretching mode with fundamental frequency $\omega_{10} = 1514.9 \text{ cm}^{-1}$ and dissociation energy $D_0 \simeq 28\omega_{10}$ inside an infrared nanocavity with the model Hamiltonian $\hat{\mathcal{H}} = \hat{H}_M + \hat{H}_C + \hat{H}_L$, where $\hat{H}_M = \hat{T}(\mathbf{q}) + \hat{V}(\mathbf{q})$ is the bare molecular Hamiltonian. $\hat{T}(\mathbf{q})$

is the molecular kinetic energy operator and $\hat{V}(\mathbf{q})$ is the potential energy with vibrational eigenstates $|\nu\rangle$, where ν is the vibrational quantum number. $\hat{H}_C = (\hat{p}^2 + \omega_c^2 \hat{x}^2)/2 + \sqrt{2\omega_c}\mathcal{E}_0\hat{d}(\mathbf{q})\hat{x}$ is the bare cavity and light-matter interaction Hamiltonians, where $\hat{d}(\mathbf{q})$ is the dipole moment function, ω_c is the cavity frequency and $\mathcal{E}_0 \equiv \lambda_g\omega_c/d_{10}$, with $d_{10} = \langle\nu = 1|\hat{d}(\mathbf{q})|\nu = 0\rangle$. The ratio $\lambda_g = g/\omega_c$ is the conventional light-matter strength parameter [43], with $g = d_{10}\mathcal{E}_0$.

We consider two scenarios for the driving Hamiltonian $\hat{H}_L$: (i) the molecule is driven by the cw laser as $\hat{H}_L^{(m)} = -\vec{d}(\mathbf{q}) \cdot \vec{\mathcal{E}}_L \sin(\omega_L t)\hat{\mathbf{e}}$, where the electric field $\vec{\mathcal{E}}_L$ is polarized along $\hat{\mathbf{e}}$; (ii) the cavity field is pumped by the laser as $\hat{H}_L^{(c)} = F_0(\hat{a}^\dagger + \hat{a}) \sin(\omega_L t)\hat{\mathbf{e}}$, where $\hat{a}$ ($\hat{a}^\dagger$) is the annihilation (creation) operator of the cavity mode and $|F_0|^2$ is proportional to the photon flux injected into the nanocavity. These two intracavity cases are illustrated in Fig. 1(b). For each case, the system is initially in the polariton ground state before the laser is turned on, and fully resonant conditions are assumed ($\omega_c = \omega_{10} = \omega_L$). Dissociation probabilities are computed by solving the time-dependent Schrödinger equation in coordinate space using the numerically exact multiconfiguration time-dependent Hartree (MCTDH) method [44–47] (additional calculation details are given in SM).

Figure 2 shows the evolution of the mean vibrational energy $\langle\hat{H}_M\rangle$ of the CS_2 stretching mode under different driving conditions, for a cavity in strong coupling ($\lambda_g = 0.02$). The free-space (solid gray line) and molecule-driving (solid black line) dynamics are similar and oscillate with the same frequency at short times ($t < 500 \text{ fs}$). This oscillation is associated with the Rabi frequency of the laser driving term. At long times ($t > 500 \text{ fs}$), the intracavity vibrational energy stabilizes faster than in free space when the molecule is driven by the laser, because the photonic mode acts as an effective reservoir for vibrational energy. This type of energy exchange does not occur in free space. For the cavity-driving scenario, the energy also oscillates but has an overall growth with time that does not stabilize under cw driving, as expected for a driven harmonic oscillator [48,49]. The energy needed to reach the same value of $\langle\hat{H}_M\rangle$ is two orders of magnitude smaller when the cavity field is driven, relative to the molecule-driving scenario. This suggests that equivalent bond-breaking probabilities can be achieved by injecting photons into the cavity with much lower intensities than by directly pumping the molecule.

To assess whether these results can be explained classically, we define the effective mean-field light-matter interaction term $\hat{H}_L^{(\text{MF})} = \hat{H}_L^{(m)} + \sqrt{2\omega_c}\mathcal{E}_0\hat{d}(\mathbf{q})\langle\hat{x}(t)\rangle_m$, which describes the molecule driven by the laser and the classical cavity quadrature $\langle\hat{x}(t)\rangle_m$. The expectation value $\langle\hat{x}(t)\rangle_m$ is computed from the total polariton wave function. The mean field evolution of $\langle\hat{H}_M\rangle$ (dashed green line in

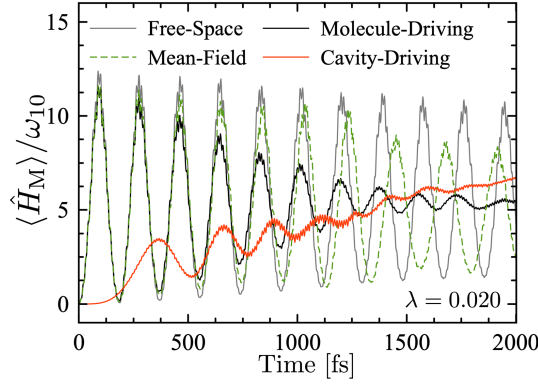


FIG. 2. Ladder-climbing dynamics. Mean vibrational energy $\langle \hat{H}_M \rangle$ as a function of time for free space, mean field, and intracavity scenarios. For intracavity cases, the light-matter coupling strength is $\lambda_g = 0.02$. Molecule-driving scenarios correspond to the driving energy $E_D \sim 0.055$ aJ and for cavity-driving $E_D \sim 0.0025$ aJ. Energy in units of fundamental vibrational frequency ω_{10} . The values of E_D are chosen to give the same long-time photodissociation probabilities in each driving scenario.

Fig. 2) is similar to the free-space evolution, because an effective laser cannot describe the energy redistribution to photonic degrees of freedom, suppressing the fast stabilization observed when the cavity field is fully quantum. This is evidence that vibrational ladder climbing is strongly influenced by quantum effects of the cavity. Additional mean-field results are given in Appendix A.

Figure 3 shows the long-time ($t = 1.5$ ps) intracavity photodissociation probability P_{diss} of CS_2 as a function of driving energy E_D and coupling strength λ_g . The driving energy is computed as $E_D = \sqrt{\langle \hat{H}_L^2 \rangle}$ both in the molecule and cavity-driving scenarios (see additional details in Supplemental Material [50]). The behavior of P_{diss} is smooth across the parameter space but a dissociation threshold can be identified as a function of driving energy for a given value of λ_g . For molecule driving [Fig. 3(a)], there is a well-defined $\lambda_g \rightarrow 0$ (free space) threshold at $E_D \approx 0.06$ aJ, which corresponds to an equivalent laser intensity of $\sim 10^{13}$ W/cm². As the coupling strength λ_g increases, this threshold intensity is reduced by an order of magnitude. Far beyond the threshold, P_{diss} becomes independent of λ_g . In this regime, the intensity is strong enough to overcome anharmonic blockade effects. In the cavity-driving scenario [Fig. 3(b)], there is also a well-defined dissociation threshold with respect to the driving energy at $\lambda_g = 0.005$ ($E_D \approx 0.003$ aJ, equivalent to a laser intensity of $\sim 10^{10}$ W/cm²). The driving energy threshold is decreased rapidly as λ_g increases. In general, the thresholds are at least an order of magnitude lower than the molecule-driving case, which can be equivalent to a reduction of up to three orders of magnitude in laser intensity. These results show that ladder climbing along vibrational polariton levels

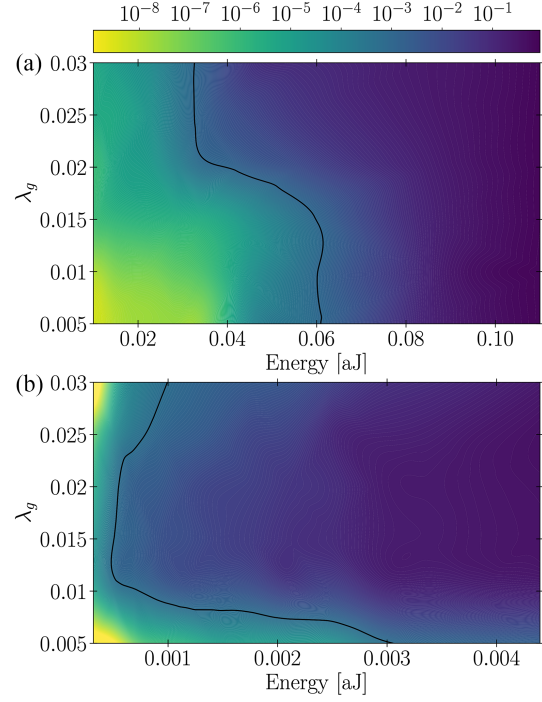


FIG. 3. Dissociation probability maps. (a) P_{diss} as a function of driving energy and coupling strength λ_g for the molecule-driving scenario at $t = 1.5$ ps; (b) same as in panel (a) for the cavity-driving scenario. Dissociation threshold contours for $P_{\text{diss}}^{(\text{th})} = 10^{-3}$ are shown (black solid lines). Energy in attojoules (aJ). For reference 0.01 aJ $\equiv 23$ THz.

is qualitatively different from conventional vibrational ladder climbing in free space.

Figure 4(a) shows the evolution of P_{diss} for an intracavity molecule directly driven with different strengths E_D (solid lines). The system is in strong coupling with $\lambda_g = 0.025$. For comparison, results for free space at the same driving energies are shown (dashed lines). For $E_D = 0.03$ aJ, the system is around the dissociation threshold in Fig. 3(a). In comparison with free space, P_{diss} increases more slowly than free space initially, but after a few hundred femtoseconds it becomes higher than free space, reaching values 10^4 times higher at long times ($t > 1.5$ ps). Doubling the driving energy ($E_D = 0.06$ aJ) gives qualitatively similar comparisons between intracavity and free space, but there are still significant longtime differences in P_{diss} . The slow increase in photodissociation probability before the activation time (black arrows) results from the light-matter interaction strength, and becomes faster for larger λ_g (see additional details in Appendix B). For high enough driving energy, there are essentially no differences between the intracavity and free space photodissociation dynamics ($E_D = 0.11$ aJ). We attribute this behavior to the laser intensity being strong enough to overcome anharmonic blockade effects via multiphoton excitations.

Figure 4(b) shows the evolution of P_{diss} for the cavity-driving scenario in strong coupling ($\lambda_g = 0.025$) subject to

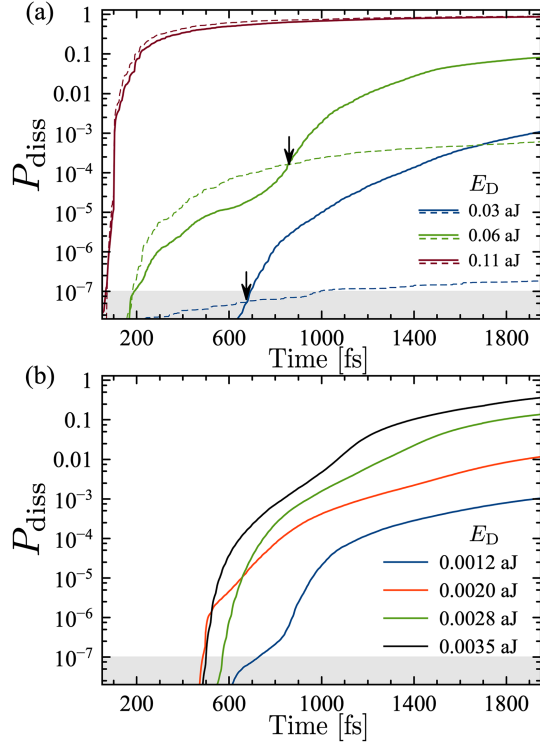


FIG. 4. Time-dependent dissociation probability. Photodissociation probability for different driving energies E_D in aJ for (a) the molecule-driving and (b) the cavity-driving scenarios. Dashed lines for the corresponding color of laser intensity represent the free-space scenario. In both panels $\lambda_g = 0.025$. The shaded region corresponds to the numerical error threshold.

different driving energies E_D above threshold. As expected, bond-breaking probabilities increase as the driving energy grows and more photons are injected into the nanocavity. Similar to the molecule-driving scenario, P_{diss} initially increases slowly due to the light-matter interaction timescale dictated by the Rabi splitting. This reveals how effectively the infrared vacuum enhances the vibrational ladder-climbing mechanism in a nanocavity, even with driving energies much lower than in the molecule-driving case.

In summary, we showed that infrared laser photodissociation of single molecules has fundamentally different dynamics inside a cavity than in free space, due to vacuum-assisted ladder-climbing transitions that occur in vibrational strong coupling. The details of the photodissociation process, such as the activation time and dissociation yield, depend on whether the laser drives directly the vibrational degrees of freedom or the cavity quadrature. In both cases, less laser energy is required to achieve a target dissociation outcome than in free space. This behavior resembles the trends found when comparing free-space infrared photodissociation of diatomic and polyatomic molecules, where the higher density of states and couplings of polyatomic molecules facilitate vibrational ladder-climbing [15–17]. In our picture, the photonic mode can be seen as an auxiliary mode that facilitates the activation of the molecular mode.

This vacuum-assisted molecular activation is a purely quantum mechanical effect that cannot be captured with a semiclassical mean-field treatment of the problem (see details in Appendix A). We can expect that cavity-induced enhancements could be less effective for large polyatomic molecules, due to their quasi-continuum nature of the vibrational spectrum arising from intramolecular vibrational couplings, which naturally suppress anharmonic blockade effects. The competition between internal vibrational processes and cavity vacuum effects remains to be explored.

The treatment used here is based on the multipolar formulation of cavity QED in strong coupling [38,51,52], which for this photodissociation problem gives equivalent results to an alternative treatment based on the Pauli-Fierz model [53,54] that includes the dipole self-energy term. Specific model comparisons are given in Appendix C. Theory improvements would require an accurate treatment of the photonic modes of nanocavities, including the effects of photon loss and dispersion within a macroscopic QED formulation [55–57]. Since we solve the time-dependent Schrödinger equation for unitary dynamics, in principle our results are strictly valid for timescales shorter than the intrinsic relaxation times. Current photon lifetimes in nanocavities are on the order of ~ 100 fs [58,59], which can be shorter than the calculated dissociation timescales. In the presence of photon loss, our conclusions should hold by compensating photon decay with higher laser intensity or light-matter coupling strength. Simpler Lindblad quantum master equations in coordinate space could also be formulated to describe dissociation in multimode dissipative cavities using available numerical methodologies [60], which is subject for future work.

Infrared cavity photodissociation can be potentially simpler to implement in experiments compared to UV photodissociation because it involves only the ground electronic state, which has a well-known dissociation energy. Currently, infrared nanocavities are being developed for reaching exotic regimes of light-matter interaction, such as ultrastrong coupling with a few vibrational dipoles [35]. Reaching strong coupling at the single-molecule level at mid-infrared wavelengths has yet to be demonstrated, and our results demonstrate that photochemistry can be fundamentally different in this limit. In addition, current experimental methodologies could be used to isolate both driving scenarios. For the molecule-driving case, probe nanotips could guide photons within the nanocavity gap to directly drive the molecules, and similar approaches can be implemented to inject photons solely into the cavity by localizing the nanotip within the metallic nanostructure far from molecules [25,59]. Our Letter can significantly contribute to the design of new chemical control mechanisms that exploit purely quantum effects to reduce energy consumption in photodissociation, as well as to the understanding of how photonic degrees of freedom influence the barrier crossing dynamics in molecular systems.

Acknowledgments—We acknowledge fruitful discussions with Johannes Schachenmayer and Blake Simpkins. J. F. T. is supported by ANID-Fondecyt Iniciación No. 11230679 and “Núcleo de Investigación No. 7 UCN-VRIDT 076/2020, Núcleo de modelación y simulación científica (NMSC).” F. H. is supported by ANID through grants Fondecyt Regular No. 1221420 and Millennium Science Initiative Program No. ICN17_012.

Data availability—The data that support the findings of this article are publicly available [61].

- [1] W. S. Warren, H. Rabitz, and M. Dahleh, *Science* **259**, 1581 (1993).
- [2] J. Zhang, D. G. Imre, and J. H. Frederick, *J. Phys. Chem.* **93**, 1840 (1989).
- [3] F. F. Crim, *Science* **249**, 1387 (1990).
- [4] N. Elghobashi, P. Krause, J. Manz, and M. Oppel, *Phys. Chem. Chem. Phys.* **5**, 4806 (2003).
- [5] R. Duperrex and H. van den Bergh, *J. Chem. Phys.* **71**, 3613 (1979).
- [6] D. S. Bomse, R. L. Woodin, and J. L. Beauchamp, *J. Am. Chem. Soc.* **101**, 5503 (1979).
- [7] E. de Lima, E. Rosado, L. Castelano, and R. Egydio de Carvalho, *Phys. Lett. A* **378**, 2657 (2014).
- [8] S. Chelkowski, A. D. Bandrauk, and P. B. Corkum, *Phys. Rev. Lett.* **65**, 2355 (1990).
- [9] J. T. Lin, T. L. Lai, D. S. Chu, and T. F. Jiang, *J. Phys. B* **31**, L117 (1998).
- [10] J.-M. Yuan and W.-K. Liu, *Phys. Rev. A* **57**, 1992 (1998).
- [11] G. Marcus, A. Zigler, and L. Friedland, *Europhys. Lett.* **74**, 43 (2006).
- [12] D. Noid and J. Stine, *Chem. Phys. Lett.* **65**, 153 (1979).
- [13] M. Kaluža, J. T. Muckerman, P. Gross, and H. Rabitz, *J. Chem. Phys.* **100**, 4211 (1994).
- [14] M. Korolkov, J. Manz, and G. Paramonov, *Chem. Phys.* **217**, 341 (1997).
- [15] M. Quack, *J. Chem. Phys.* **69**, 1282 (1978).
- [16] B. Carmeli and A. Nitzan, *J. Chem. Phys.* **72**, 2070 (1980).
- [17] V. S. Letokhov and A. A. Makarov, *Sov. Phys. Usp.* **24**, 366 (1981).
- [18] J. P. Long and B. S. Simpkins, *ACS Photonics* **2**, 130 (2015).
- [19] B. S. Simpkins, K. P. Fears, W. J. Dressick, B. T. Spann, A. D. Dunkelberger, and J. C. Owrutsky, *ACS Photonics* **2**, 1460 (2015).
- [20] J. George, T. Chervy, A. Shalabney, E. Devaux, H. Hiura, C. Genet, and T. W. Ebbesen, *Phys. Rev. Lett.* **117**, 153601 (2016).
- [21] F. Herrera and J. Owrutsky, *J. Chem. Phys.* **152**, 100902 (2020).
- [22] J. George, A. Shalabney, J. A. Hutchison, C. Genet, and T. W. Ebbesen, *J. Phys. Chem. Lett.* **6**, 1027 (2015).
- [23] A. D. Dunkelberger, B. T. Spann, K. P. Fears, B. S. Simpkins, and J. C. Owrutsky, *Nat. Commun.* **7**, 13504 (2016).
- [24] B. Xiang, R. F. Ribeiro, Y. Li, A. D. Dunkelberger, B. B. Simpkins, J. Yuen-Zhou, and W. Xiong, *Sci. Adv.* **5**, eaax5196 (2019).
- [25] B. Metzger, E. Muller, J. Nishida, B. Pollard, M. Hentschel, and M. B. Raschke, *Phys. Rev. Lett.* **123**, 153001 (2019).
- [26] E. A. Muller, B. Pollard, H. A. Bechtel, R. Adato, D. Etezadi, H. Altug, and M. B. Raschke, *ACS Photonics* **5**, 3594 (2018).
- [27] J. F. Triana, M. Arias, J. Nishida, E. A. Muller, R. Wilcken, S. C. Johnson, A. Delgado, M. B. Raschke, and F. Herrera, *J. Chem. Phys.* **156**, 124110 (2022).
- [28] J. Wang, K. Yu, Y. Yang, G. V. Hartland, J. E. Sader, and G. P. Wang, *Nat. Commun.* **10**, 1527 (2019).
- [29] K. S. Menghrajani, H. A. Fernandez, G. R. Nash, and W. L. Barnes, *Adv. Opt. Mater.* **7**, 1900403 (2019).
- [30] A. Thomas, L. Lethuillier-Karl, K. Nagarajan, R. M. A. Vergauwe, J. George, T. Chervy, A. Shalabney, E. Devaux, C. Genet, J. Moran, and T. W. Ebbesen, *Science* **363**, 615 (2019).
- [31] J. Lather and J. George, *J. Phys. Chem. Lett.* **12**, 379 (2021).
- [32] W. Ahn, J. F. Triana, F. Recabal, F. Herrera, and B. S. Simpkins, *Science* **380**, 1165 (2023).
- [33] Z. T. Brawley, S. Pannir-Sivajothi, J. E. Yim, Y. R. Poh, J. Yuen-Zhou, and M. Sheldon, *Nat. Chem.* **17**, 439 (2025).
- [34] F. Bell, L. Jakob, C. Todd, I. Lohia, Y. Roh, R. Arul, and J. J. Baumberg, *Phys. Rev. Lett.* **135**, 076901 (2025).
- [35] D. Yoo, F. de León-Pérez, M. Pelton, I.-H. Lee, D. A. Mohr, M. B. Raschke, J. D. Caldwell, L. Martín-Moreno, and S.-H. Oh, *Nat. Photonics* **15**, 125 (2021).
- [36] R. Arul, K. Menghrajani, M. S. Rider, R. Chikkaraddy, W. L. Barnes, and J. J. Baumberg, *Phys. Rev. Lett.* **131**, 126902 (2023).
- [37] E. Suyabatmaz, G. J. R. Aroeira, and R. F. Ribeiro, *J. Phys. Chem. Lett.* **16**, 7530 (2025).
- [38] J. F. Triana and F. Herrera, *J. Chem. Phys.* **162**, 134103 (2025).
- [39] A. Kadyan, A. Shaji, and J. George, *J. Phys. Chem. Lett.* **12**, 4313 (2021).
- [40] S. Mukamel and J. Jortner, *J. Chem. Phys.* **65**, 5204 (1976).
- [41] R. J. Levis and M. J. DeWitt, *J. Phys. Chem. A* **103**, 6493 (1999).
- [42] F. Herrera and F. C. Spano, *Phys. Rev. Lett.* **118**, 223601 (2017).
- [43] W. Demtröder, *Molecular Physics: Theoretical Principles and Experimental Methods* (John Wiley & Sons, New York, 2008).
- [44] M. Beck, A. Jackle, G. Worth, and H.-D. Meyer, *Phys. Rep.* **324**, 1 (2000).
- [45] U. Manthe, H. Meyer, and L. S. Cederbaum, *J. Chem. Phys.* **97**, 3199 (1992).
- [46] H.-D. Meyer, U. Manthe, and L. Cederbaum, *Chem. Phys. Lett.* **165**, 73 (1990).
- [47] H.-D. Worth, G. Beck, M. Jäckle, and A. Meyer, The MCTDH Package, (2021), <http://mctdh.uni-hd.de/>.
- [48] P. Hänggi, T. Dittrich, G. Ingold, B. Kramer, G. Schön, and W. Zwerger, *Quantum Transport and Dissipation* (Wiley-Vch, Weinheim, 1998), Vol. 3.
- [49] P. C. López Vázquez, *Phys. Rev. A* **98**, 042128 (2018).
- [50] See Supplemental Material at <http://link.aps.org/supplemental/10.1103/s2zc-6lxs> for numerical and computational details, and driving energy derivations.

- [51] F. J. Hernández and F. Herrera, *J. Chem. Phys.* **151**, 144116 (2019).
- [52] J. F. Triana, F. J. Hernández, and F. Herrera, *J. Chem. Phys.* **152**, 234111 (2020).
- [53] J. Flick, M. Ruggenthaler, H. Appel, and A. Rubio, *Proc. Natl. Acad. Sci. U.S.A.* **114**, 3026 (2017).
- [54] M. Ruggenthaler, D. Sidler, and A. Rubio, *Chem. Rev.* **123**, 11191 (2023).
- [55] B. Candelas, N. Zabala, P. Koval, A. Babaze, D. Sánchez-Portal, and J. Aizpurua, *J. Chem. Phys.* **161**, 044707 (2024).
- [56] L.-Y. Hsu, *J. Phys. Chem. Lett.* **16**, 1604 (2025).
- [57] A. S. Rema, A. E. R. López, and F. Herrera, *arXiv:2509.03428*.
- [58] M. Autore, P. Li, I. Dolado, F. J. Alfaro-Mozaz, R. Esteban, A. Atxabal, F. Casanova, L. E. Hueso, P. Alonso-González, J. Aizpurua, A. Y. Nikitin, S. Vélez, and R. Hillenbrand, *Light Sci. Appl.* **7**, 17172 (2018).
- [59] R. Wilcken, J. Nishida, J. F. Triana, A. John-Herpin, H. Altug, S. Sharma, F. Herrera, and M. B. Raschke, *Proc. Natl. Acad. Sci. U.S.A.* **120**, e2220852120 (2023).
- [60] J. F. Triana and F. Herrera, *J. Chem. Phys.* **157**, 194104 (2022).
- [61] J. F. Triana and F. Herrera, Enhancing infrared laser dissociation of molecules with the electromagnetic vacuum—dataset, [10.5281/zenodo.20298230](https://zenodo.org/record/20298230) (2026).
- [62] L. Garziano, A. Settineri, O. Di Stefano, S. Savasta, and F. Nori, *Phys. Rev. A* **102**, 023718 (2020).
- [63] M. Göppert-Mayer, *Ann. Phys. (Berlin)* **401**, 273 (1931).
- [64] C. Schäfer, M. Ruggenthaler, and A. Rubio, *Phys. Rev. A* **98**, 043801 (2018).
- [65] V. Rokaj, D. M. Welakuh, M. Ruggenthaler, and A. Rubio, *J. Phys. B* **51**, 034005 (2018).
- [66] P. Forn-Díaz, J. Lisenfeld, D. Marcos, J. J. García-Ripoll, E. Solano, C. J. P. M. Harmans, and J. E. Mooij, *Phys. Rev. Lett.* **105**, 237001 (2010).
- [67] S.-P. Wang, G.-Q. Zhang, Y. Wang, Z. Chen, T. Li, J. S. Tsai, S.-Y. Zhu, and J. Q. You, *Phys. Rev. Appl.* **13**, 054063 (2020).
- [68] A. Semenov and A. Nitzan, *J. Chem. Phys.* **150**, 174122 (2019).
- [69] Y.-C. Wei and L.-Y. Hsu, *J. Phys. Chem. Lett.* **13**, 9695 (2022).
- [70] Y.-C. Wei and L.-Y. Hsu, *J. Phys. Chem. Lett.* **14**, 2395 (2023).
- [71] E. A. Power and T. Thirunamachandran, *Phys. Rev. A* **28**, 2649 (1983).
- [72] D. L. Andrews, G. A. Jones, A. Salam, and R. G. Woolley, *J. Chem. Phys.* **148**, 040901 (2018).
- [73] J. Feist, A. I. Fernández-Domínguez, and F. J. García-Vidal, *Nanophotonics* **10**, 477 (2020).
- [74] D. Craig and T. Thirunamachandran, *Molecular Quantum Electrodynamics: An Introduction to Radiation-molecule Interactions*, Dover Books on Chemistry Series (Dover Publications, New York, 1998).
- [75] S. Buhmann, *Dispersion Forces I: Macroscopic Quantum Electrodynamics and Ground-State Casimir, Casimir-Polder and van der Waals Forces*, Springer Tracts in Modern Physics (Springer, Berlin, Heidelberg, 2013).

End Matter

Appendix A: Quantum dynamics simulations in the mean-field approximation—We perform a mean-field analysis to understand how the quantum effects of the photonic mode influence the modifications of the infrared photodissociation probability. The mean-field Hamiltonian associated with molecule-driving scenario $\hat{\mathcal{H}}_{\text{MF}}^{(\text{mol})}$ is given by

$$\hat{\mathcal{H}}_{\text{MF}}^{(\text{mol})} = \hat{H}_{\text{M}} + \hat{H}_{\text{L}} + \sqrt{2\omega_{\text{c}}}\mathcal{E}_0\hat{d}(q)\langle\hat{x}(t)\rangle_{\text{M}}, \quad (\text{A1})$$

where $\langle\hat{x}(t)\rangle_{\text{M}}$ is the expectation value of coordinate x for the photonic mode, which is calculated with the total wave function that results from the time-dependent Schrödinger equation solution with the light-matter Hamiltonian $\hat{\mathcal{H}}$. The mean-field approximation can be seen as a modulation of the cw laser; i.e., the molecule is driven by an effective laser that results in an effective driven Hamiltonian

$$\hat{H}_{\text{L-M}}^{(\text{eff})} = -\hat{d}(q)\left[\hat{\mathcal{E}}_{\text{L}}\sin(\omega_{\text{c}}t) - \sqrt{2\omega_{\text{c}}}\mathcal{E}_0\langle\hat{x}(t)\rangle_{\text{M}}\right].$$

Similarly, the mean-field Hamiltonian for the cavity-driving scenario reads

$$\hat{\mathcal{H}}_{\text{MF}}^{(\text{cav})} = \hat{H}_{\text{M}} + \sqrt{2\omega_{\text{c}}}\mathcal{E}_0\hat{d}(q)\langle\hat{x}(t)\rangle_{\text{C}}, \quad (\text{A2})$$

where $\langle\hat{x}(t)\rangle_{\text{C}} = \sqrt{2\omega_{\text{c}}}F_0[t\sin(\omega_{\text{c}}t)]$ is the solution of the driven cavity Hamiltonian

$$\hat{H}_{\text{CL}} = -\frac{1}{2}\frac{\partial^2}{\partial\hat{x}^2} + \frac{1}{2}\omega_{\text{c}}\hat{x}^2 + \sqrt{2\omega_{\text{c}}}F_0\hat{x}\sin(\omega_{\text{c}}t), \quad (\text{A3})$$

and the effective laser Hamiltonian in this scenario is given by $\hat{H}_{\text{L-C}}^{(\text{eff})} = \sqrt{2\omega_{\text{c}}}\mathcal{E}_0\hat{d}(q)\langle\hat{x}(t)\rangle_{\text{C}}$.

Figure 5(a) show the expectation values of the effective driven Hamiltonian for the molecule and cavity-driving scenario, respectively. The most notable difference refers to the laser field strength in the cavity-driving scenario, which, on average, is approximately one order of magnitude lower than in the molecule-driving case. The functional shape of $\hat{H}_{\text{L-M}}^{(\text{eff})}$ in Fig. 5(a) varies along time, but its amplitude does not increase continuously, in comparison with the cavity-driving case that depends linearly on time.

Figure 5(b) shows the normalized molecular mean energy $\langle\hat{H}_{\text{M}}\rangle/\omega_{10}$ for the mean-field (dashed line) and quantum (solid line) scenarios of the molecule-driving scenario. The mean molecular energy in the mean-field approximation tends to stabilize more slowly than the

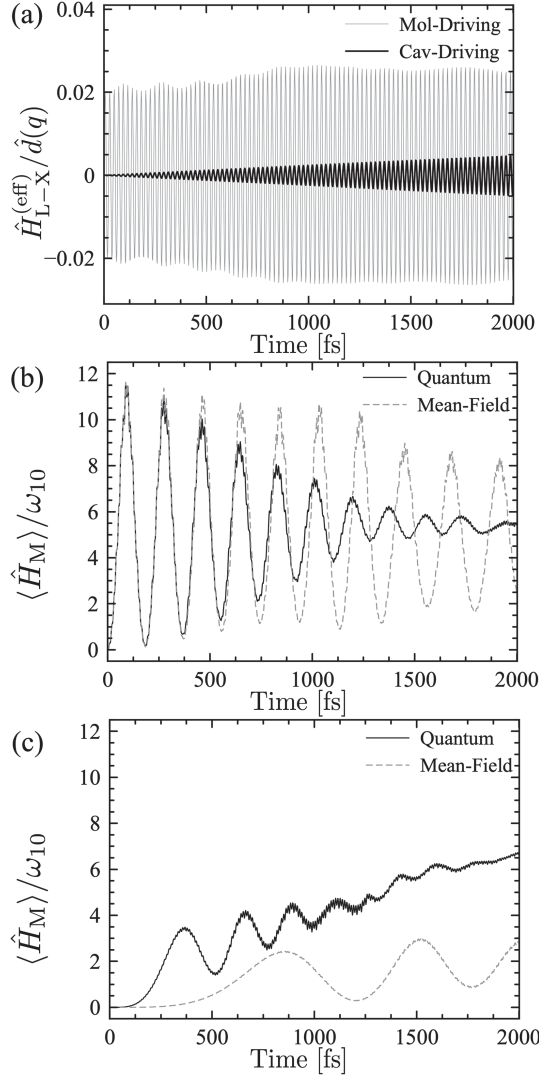


FIG. 5. Mean-field approximation. (a) Expectation value of the effective driven Hamiltonian $\hat{H}_{L-X}^{(eff)}/\hat{d}(q)$ as a function of time for molecule-driving (gray line) and cavity-driving (black line) scenarios, with $X = M, C$. Vibrational mean energy as a function of time for (b) molecule-driving and (c) cavity-driving scenarios. In all cases $\lambda_g = 0.02$ and molecule-driving scenario correspond to $E_D \sim 0.055$ aJ and for cavity-driving $E_D \sim 0.0025$ aJ.

intracavity scenario, similar to the free-space scenario described in the main text. However, the molecular mean energy in the mean-field cavity-driving case increases as the effective driving energy increases at a small rate in comparison with the complete quantum scenario, as Fig. 5(c) shows. The latter is because the number of photons injected into the cavity interacts more efficiently than by driving the molecule directly.

Appendix B: Dissociation Probabilities as a function of λ_g —Figure 6 shows P_{diss} for different coupling strengths and a driving energy $E_D = 0.03$ aJ. In free space (dashed line), the dissociation probability is very

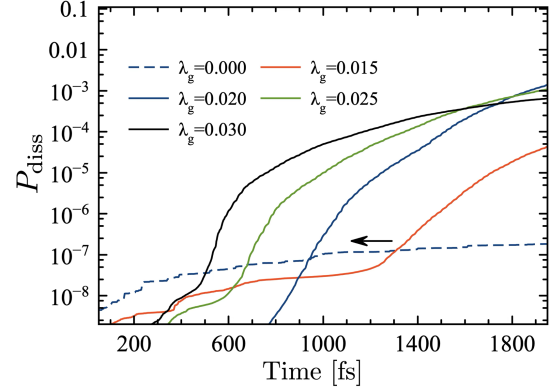


FIG. 6. Time-dependent dissociation probability. Intracavity dissociation probabilities for $E_D = 0.03$ aJ and different light-matter couplings λ_g . The arrow indicates the shift in the activation time for the intracavity photodissociation probability as λ_g increases.

low and remains almost constant around 10^{-6} over the entire propagation time. In contrast, intracavity dissociation probabilities (solid lines) increase by several orders of magnitude and exhibit a remarkable monotonic coupling-dependent trend, as illustrated by the arrow. This monotonic behavior reveals that the time at which P_{diss} begins to rise with respect to free space, which we call activation time, is determined by the value of λ_g . The latter implies that intracavity photodissociation processes are governed by light-matter interactions instead of initial polariton blockade effects [see Fig. 1(b)], which can occur in the weak-field limit.

Appendix C: Comparison with Pauli-Fierz model—The Pauli-Fierz model for a single vibrational mode coupled to a single field mode can be written as [53,54]

$$\hat{\mathcal{H}}_{\text{PF}} = \hat{H}_M + \hat{H}_C + \hat{H}_L + \frac{\mathcal{E}_0^2}{\omega_c} [\hat{d}(q)]^2, \quad (\text{C1})$$

where $\hat{H}_M$, $\hat{H}_C$ and $\hat{H}_L$ are the molecular, photonic, and driving Hamiltonians defined in the main text, respectively. This model is derived from a mode-truncated minimal coupling formulation in the electric dipole approximation through a mode-truncated PZW unitary or Göppert-Mayer transformation [62,63]. The last term in Eq. (C1) is known as the dipole self-energy (DSE) term [64,65], and increases linearly with vacuum field fluctuations $\mathcal{E}_0^2$. This dipole self-energy term could become a leading contribution as the light-matter coupling increases, giving rise to physical effects that have not been observed in current USC implementations [66,67]. This term is controversial and has been discussed in recent works [68–70]. In contrast, the light-matter Hamiltonian model used in this work is based on the multipolar formulation of cavity QED in strong coupling [38,51,52],

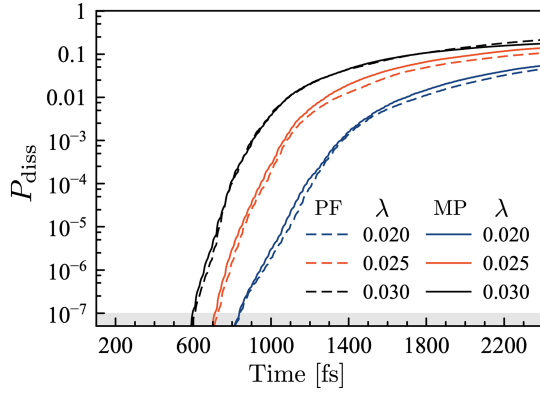


FIG. 7. Pauli-Fierz vs multipolar QED dissociation probability. Dissociation probability for different values of λ_g with $I_0 = 8.0 \times 10^{12} \text{ W/cm}^2$ ($E_D = 0.055 \text{ aJ}$), for the molecule-driving scenario. Solid lines represent P_{diss} calculated without DSE, and dashed lines correspond to dissociation probabilities found with the Pauli-Fierz Hamiltonian.

which is equivalent to the Hamiltonian that arises from the minimal coupling form via a Power-Zienau-Woolley (PZW) unitary transformation. We point out that the DSE is not included in conventional multipolar formulations of QED, because it can be formally absorbed into the potential energy curve [71–75].

Figure 7 shows photodissociation probabilities of C-S stretch mode of the CS_2 molecule for different values of coupling strength λ_g and driving energy $E_D = 0.055 \text{ aJ}$. Photodissociation probabilities computed with the Pauli-Fierz model in Eq. (C1) in the molecule-driving scenario (dashed lines) give qualitatively similar predictions as a function of time as those calculated using the Hamiltonian $\hat{\mathcal{H}}$ derived from multipolar QED formulation (solid lines). Hence, we highlight that regardless of the microscopic origin of the light-matter Hamiltonian used by different authors, the physics of our dissociation results holds in both mode-truncated models.